

EEE 419/591

Summ 2026

Midterm Project – Big Data Feature Selection

Introduction

When working with real-world machine learning systems, data engineers routinely encounter datasets with massive dimensions. Let's consider a scenario where a dataset contains a very high feature space (e.g., 10,000 features) spread across tens of thousands of observation rows. Standard machine learning algorithms generally expect a unified 2D matrix to be loaded completely into the computer's volatile memory (RAM). However, when a dataset scales into gigabytes or terabytes, attempting an "in-core" processing loop can immediately deplete your hardware resources and trigger Out-of-Memory system crashes.

To circumvent this, suboptimal algorithms are usually deployed. These algorithms depend on only a portion of the dataset. This portion could be either a row-wise portion of the dataset, or a column-wise portion. The former is used when the dataset cannot be loaded into the memory because of the massive amount of dataset points collected. The latter, on the contrary, is more valuable when the feature space is massive, which is the scope of this project. The dataset provided for this project is, thus, stored in a format called "vertically partitioned" format that allows computationally efficient loading of only some of the features. Since the hurdle in this project is not the number of dataset points, loading all dataset points is allowed and even encouraged. This is because loading only a few of the dataset points in a vertically partitioned dataset would not be time efficient even if it is feasible in terms of the amount of memory it takes.

Many techniques have been proposed to select the features efficiently without loading the entire dataset into the RAM. Such techniques include, but are not limited to, Principal Component Analysis (PCA), correlation coefficients, random selection, manual evaluation or even a combination of these. The fundamental goal of this project is to learn how to use techniques in the literature to interact with the vertically partitioned dataset you are given to train and test a classifier using a computer that has hardware constraints that limit the amount of data you can load into the RAM simultaneously.

Problem Statement

In this project, you are asked to write a complete Python feature selection and multi-class classification evaluation code for the vertically partitioned dataset provided. You are allowed to use any idea(s) you deem fit to achieve this goal as long as it does not involve loading the entire dataset into the RAM all at once.

Dataset Properties

The dataset provided has the following parameters:

- Number of Features: 10,000
- Number of training dataset points: 70,000
- Number of testing dataset points: 30,000
- Number of classes: 100 (labelled 0-99)

Dataset Structure

Each of the training and testing portions is further split into 10 chunks, each chunk spans 1,000 consecutive features. Each chunk is stored in a separate file named according to its index bounds and whose name ends in “*.parquet*” extension. Further, the label (i.e. class) column for each is provided in a separate file named “*labels_param.parquet*”.

Dataset Acquiring

You have two options to acquire the dataset for this project:

1. Download the pre-generated dataset from the following link (~5.5GB):
<https://www.dropbox.com/scl/fo/7wb8ae336d8atq0yyb38z/ADm81ofT17J45IWhf4eIVi8?rlkey=4u6jfohav37qupp1cde7gj3ky&st=m7v7n2fm&dl=0>
2. Run the provided dataset generation script *dataset_gen.py* on your machine without changing any parameters (10-15 mins to run). Do not modify the generation code or parameters at all. Otherwise, your script will run on a different dataset than what the grader is expecting, resulting in a failed evaluation.

After the dataset is saved on your computer, you can load any of its files in your code either via the example commands provided to you in the project *midterm_project.py* template, or any other commands you deem fit. For the provided commands to work, make sure that your dataset directory structure looks like this:

Parent_directory < where your to-be-developed project file is located

```
| midterm_dataset/  
| | — train/  
| | | — labels_param.parquet <-- Contains 'class_label' & encoded metadata  
| | | — features_0_1000.parquet <-- Feature Chunk 1 (Columns 0-999)  
| | | — features_1000_2000.parquet <-- Feature Chunk 2 (Columns 1000-1999)  
| | | ... <-- Remaining Feature Chunks  
| | — test/  
| | | — labels_param.parquet <-- Contains 'class_label' & encoded metadata  
| | | — features_0_1000.parquet <-- Feature Chunk 1 (Columns 0-999)  
| | | — features_1000_2000.parquet <-- Feature Chunk 2 (Columns 1000-1999)  
| | | ... <-- Remaining Feature Chunks
```

Technical Requirements

Your code must meet the following requirements:

1. **RAM Constraint**: Your solution **MUST NOT** load the entire dataset into the RAM all at once. At most, your code may only hold 25% of the total dataset features in active memory at any given moment. Students who submit a code that loads the entire dataset in the RAM will automatically receive a grade of zero.
2. **Classification**: Your code must be an ML-based algorithm. Non-ML-based codes will not be accepted.
3. **Runtime Limit**: Your entire script must be optimized. Codes that take more than 40 minutes to run on your own computer will not be accepted. It will typically take far less than that if run on an average computer in the market.
4. **Performance Expectations**: The classification performance does not need to be perfectly optimal. You will not be graded on the absolute accuracy score unless your result is too far (>10% lower) from the maximum accuracy reported on Ed Discussion by your peers.

Deliverables

- You must submit a single Python file named ***Midterm_Project_FirstName_LastName.py***.
- Do not submit the dataset.
- Your code must expect the dataset to be in the same format and name it is provided
- Your code must print out the final validation accuracy and the total execution time directly to the terminal.
- No written reports or documentation are needed. If you experimented with multiple machine learning algorithms or parameters that did not work as expected, do not include them in your final code submission. Only the single algorithm that successfully works is the deliverable.

Policies

Allowed:

1. The use of AI (ChatGPT...etc.), especially if you use it to:
 - Ask it to recommend specific commands that perform a task.
 - Paste parts of your non-working code along with error messages to understand why it is failing.
 - *Note: You should not use AI suggestions blindly since AI can hallucinate.*
2. Asking AI anything.
3. The use of any external resources (internet, textbooks, instructional videos, forums).
4. Discussions through the Ed Discussion board with any question/answer/code (except sharing your entire code). This is highly encouraged to deepen your understanding. Reminder that it is best practice to limit your screenshots or code shares to the commands that deliver the message you want to deliver as opposed to sharing a lot and expecting the reader to read everything and figure it out themselves.
5. Sharing final answers (e.g., your final accuracy percentage or total execution runtime). Make sure you don't create a new post for this if one has already been created.
6. Sharing ranges of parameter values you tried or recommend.
7. Sharing the name of the classification or preprocessing algorithms you plan to submit, work on, or have (not) tried and, more encouragingly, why you (not) recommend them.
8. Sharing screenshots of codes that don't work or give errors to ask your peers for feedback.
9. Sharing fundamental conceptual ideas that would work or would not work.

Not Allowed:

1. Submitting the entire code that an AI gives you without understanding what it does. AI is allowed because it helps you understand and be productive, but keep in mind that, by the end of the day, it is still a "robot". You are responsible for what you submit.
2. Seeking help from any human being outside this class (private tutors, friends, family, etc.).
3. Meeting to work on the entire project (or even a major portion of it) together. This project is an individual effort. Discussion and asking specific questions over Ed Discussion, regardless of how much information they reveal, however, is completely allowed and even encouraged.

MANDATORY – Reminder to cite your AI chat. Check the syllabus document for more details